

Activity 3 Gradient Magnitude and Gradient Direction:-

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IIQA0502

Computer Vision
SotC.

→ Comp

→ P122

→ P22

→ P32

→ P42

→ P52

After r

Pine

Res

A) Given;

$$G_x = 24$$

$$G_y = 18$$

1) Gradient Magnitude:-

Formula:-

$$G = \sqrt{G_x^2 + G_y^2}$$

$$G = \sqrt{24^2 + 18^2}$$

$$= \sqrt{900}$$

$$\boxed{G = 30}$$

2) Gradient Direction:-

Formula:-

$$\theta = \tan^{-1}\left(\frac{G_y}{G_x}\right)$$

$$\theta = \tan^{-1}\left(\frac{18}{24}\right)$$

$$\theta = \tan^{-1}(0.75)$$

$$\boxed{\therefore \theta \approx 36.87^\circ}$$

Ans:-

Gradient Magnitude = 30

Gradient Direction = 36.87°

Q2) Non-Maximum Suppression.

A) Given;

Pixel	P ₁	P ₂	P ₃	P ₄	P ₅
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Magnitude	25	50	90	60	30
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∴ Gradient direction is horizontal.

→ It keeps a Pixel only if it has a larger Gradient value than its Pixels in the Gradient direction.

Q3)

A) Given

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Rule

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Cl

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→ Compare value

* $P_1 = 25 < P_2 = 50$
(suppress)

* $P_2 = 50 < P_3 = 90$
(suppress)

* $P_3 = 90 > P_2 = 50$ and $P_4 = 60$
(keep)

* $P_4 = 60 < P_3 = 90$ (suppress)

* $P_5 = 30 < P_4 = 60$ (suppress)

After non-Max Suppression

<u>Pixel</u>	P_1	P_2	P_3	P_4	P_5
<u>Result</u>	0	0	90	0	0

∴ $P_3 = 90$ is retained

Q3) Classifying Pixels Using Double Thresholding.

A) Given:

High threshold = 100

Low threshold = 50

<u>Pixel</u>	P_1	P_2	P_3	P_4	P_5	P_6
<u>Magnitude</u>	135	90	45	120	70	20

Rules:

* Magnitude $\geq 100 \rightarrow$ Strong edge

* Magnitude $50 < \text{value} < 100 \rightarrow$ weak edge

* Magnitude $< 50 \rightarrow$ Non-edge

Classification:

* $P_1 = 135$

$135 \geq 100 \rightarrow$ Strong

* $P_2 = 90$

$50 \leq 90 < 100 \rightarrow$ weak

$$P_3 = 45$$

$45 < 50 \rightarrow \text{NON-edge}$

$$P_4 = 120$$

$120 \geq 100 \rightarrow \text{Strong}$

$$P_5 = 70$$

$50 \leq 70 < 100 \rightarrow \text{weak}$

$$P_6 = 20$$

$20 < 50 \rightarrow \text{NON edge}$

Conclusion:

Strong: P_1, P_4

Weak: P_2, P_5

NonEdges: P_3, P_6

(Q4) - Edge Tracking - by hysteresis.

A) Given:

Pixel

Magnitude

P_1	P_2	P_3	P_4	P_5	P_6	P_7
130	90	40	115	65	25	105

Thresholds:

High threshold = 100

Low threshold = 50

Step 1:- Classify the Pixels

$$P_1 = 130$$

$P_1 \geq 100 \rightarrow \text{Strong}$

$$P_2 = 90$$

$50 \leq 90 < 100 \rightarrow \text{weak}$

$$P_3 = 40$$

$P_3 < 50 \rightarrow \text{NON-edge}$

$$P_4 = 115$$

$\geq 100 \rightarrow \text{Strong}$

$$P_5 = 65$$

$$P_6 = 25$$

< 50 — non-edge.

$$P_7 = 100$$

$\geq 100 \rightarrow$ strong.

$\therefore$ Strong: P_1, P_4, P_7

Weak: P_2, P_5

Non: P_3, P_6

STEP 2: My steps

* P_2 is connected to P_1 .

* P_5 is not connected to any strong edge.

$\therefore$ A weak edge is kept only when it is connected to a strong edge.

